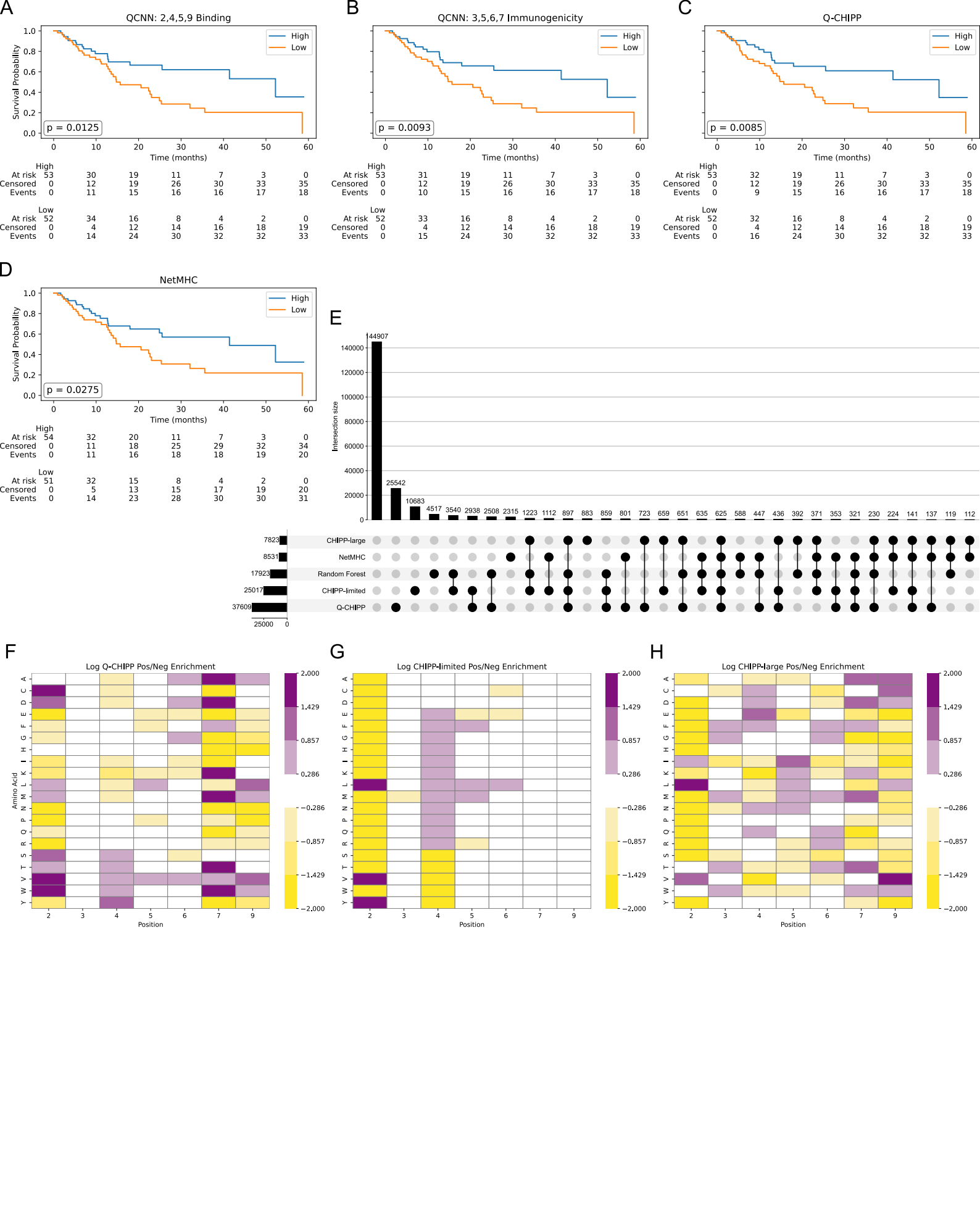
A
QCNN: 2,4,5,9 Binding
Survival Probability
Time (months)
High
Low
p = 0.0125
High
At risk 53 30 19 11 7 3 0
Censored 0 12 19 26 30 33 35
Events 0 11 15 16 16 17 18
Low
At risk 52 34 16 8 4 2 0
Censored 0 4 12 14 16 18 19
Events 0 14 24 30 32 32 33
B
QCNN: 3,5,6,7 Immunogenicity
p = 0.0093
High
At risk 53 31 19 11 7 3 0
Censored 0 12 19 26 30 33 35
Events 0 10 15 16 16 17 18
Low
At risk 52 33 16 8 4 2 0
Censored 0 4 12 14 16 18 19
Events 0 15 24 30 32 32 33
C
Q-CHIPP
p = 0.0085
High
At risk 53 32 19 11 7 3 0
Censored 0 12 19 26 30 33 35
Events 0 9 15 16 16 17 18
Low
At risk 52 32 16 8 4 2 0
Censored 0 4 12 14 16 18 19
Events 0 16 24 30 32 32 33
D
NetMHC
p = 0.0275
High
At risk 54 32 20 11 7 3 0
Censored 0 11 18 25 29 32 34
Events 0 11 16 18 18 19 20
Low
At risk 51 32 15 8 4 2 0
Censored 0 5 13 15 17 19 20
Events 0 14 23 28 30 30 31
E
Intersection size
144907
25542
10683
4517 3540 2938 2508 2315 1223 1112 897 883 859 801 723 659 651 635 625 588 447 436 392 371 353 321 230 224 141 137 119 112
7823 CHIPP-large
8531 NetMHC
17923 Random Forest
25017 CHIPP-limited
37609 Q-CHIPP
F
Log Q-CHIPP Pos/Neg Enrichment
Amino Acid
Position
G
Log CHIPP-limited Pos/Neg Enrichment
Position
H
Log CHIPP-large Pos/Neg Enrichment
Position